



A Trace inequality in Quantum Information Theory

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Introduction

In this semester, we continue our investigation into a special type of trace inequality that arise in quantum information theory and matrix analysis. This inequality plays a role in quantum theory and is connecting different entropies.

Our study focuses on generalizing a particular trace inequality involving density matrices. Specifically, we aim to advance the previous results in two main directions:

- **Finding a tighter bound:** We aim to improve the constant in the inequality, thus sharpening its behavior.
- **Relaxing the conditions:** We aim to establish the inequality under weaker assumptions, such as relaxing the unital trace condition.

The original inequality, which serves as the foundation of our work, is stated as follows: for any density matrices ρ, σ and for any $\alpha \in (1, 2]$,

$$\text{Tr} \rho (\log (\frac{\rho + \sigma}{2}) - \log \sigma) \leq \frac{1}{\alpha - 1} \text{Tr} (\sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}})^\alpha,$$

Our goal is to explore whether the bound in this inequality can be improved, and under what broader conditions it remains valid.

Notation

The following is a list of commonly used notations, adapted from [MLDS⁺13].

- **Density matrix:** ρ, σ which are positive semi-definite, trace unital matrix
- **PSD matrix:** $\mathbf{A}, \mathbf{B}$ which are positive semi-definite matrix
- **Sandwiched quasi-relative entropy:** $\tilde{Q}_\alpha(A|B) = \text{Tr} (B^{\frac{1-\alpha}{2\alpha}} A B^{\frac{1-\alpha}{2\alpha}})^\alpha$
- **Quantum relative entropy:** $D(A|B) = \text{Tr} A (\log (A + B) - \log B)$
- $\lambda_i(A)$ be the i -th largest eigenvalue of matrix A

The notations will be referred again if needed.

Tighter Bounds

Theorem (A) For density matrices ρ, σ ,

$$\text{Tr} \rho (\log (\rho + \sigma) - \log \sigma) \leq \frac{1}{\alpha - 1} \text{Tr} (\sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}})^\alpha,$$

for all $\alpha \in (1, 2]$.

Proof: Since $\text{Tr} \rho (\log (\rho + \sigma) - \log \sigma) - \log 2 = \text{Tr} \rho (\log \frac{\rho + \sigma}{2} - \log \sigma)$, we only need to show

$$\text{Tr} \rho (\log \frac{\rho + \sigma}{2} - \log \sigma) + \log 2 \leq \frac{1}{\alpha - 1} \text{Tr} (\sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}})^\alpha$$

The rest of the proof is similar to the original method. First, observe that to show the inequality, it is sufficient to show $2^{\alpha-1} \cdot \tilde{Q}_\alpha(\rho|\sigma) \leq e^{\frac{\alpha-1}{\beta-1} \tilde{Q}_\beta(\rho|\sigma)}$, for $1 < \alpha < \beta < 2$, which simplifies to

$$\tilde{Q}_\alpha(\rho|\sigma) \leq e^{\frac{\alpha-1}{\beta-1} \tilde{Q}_\beta(\rho|\sigma) - (\alpha-1) \log 2}$$

This can be shown by observing the following chain of inequalities:

$$\begin{aligned} \tilde{Q}_\alpha(\rho|\sigma) &\leq \frac{\beta - \alpha}{\beta - 1} + \frac{\alpha - 1}{\beta - 1} \tilde{Q}_\beta(\rho|\sigma) \leq 1 - (\alpha - 1) \log 2 + \frac{\alpha - 1}{\beta - 1} \tilde{Q}_\beta(\rho|\sigma) \\ &\leq e^{\frac{\alpha-1}{\beta-1} \tilde{Q}_\beta(\rho|\sigma) - (\alpha-1) \log 2} \end{aligned}$$

The second inequality is improved by using the fact that $\log 2 \leq \frac{1}{\beta-1}$ for $1 < \beta < 2$. Next, since

$$\text{Tr} \rho (\log (\rho + \sigma) - \log \sigma) \leq \frac{1}{\alpha - 1} \log 2^{\alpha-1} \cdot \tilde{Q}_\alpha(\rho|\sigma) \leq \frac{1}{\beta - 1} \tilde{Q}_\beta(\rho|\sigma)$$

for $1 < \alpha < \beta \leq 2$, by taking $\alpha \rightarrow 1$, we can complete the proof. $\square$

PSD Matrix under Normalized Condition

It is worthy to observe that for any PSD matrix A , the normalized matrix $A' = \frac{A}{\text{Tr} A}$ is a density matrix. Thus, we can investigate a natural extension of Theorem (A).

Proposition (B) For semi-positive definite, self-adjoint matrices A, B ,

$$\begin{aligned} \text{Tr} A (\log (A + B) - \log B) &\leq \frac{b^{\alpha-1}}{\alpha - 1} \text{Tr} (B^{\frac{1-\alpha}{2\alpha}} A B^{\frac{1-\alpha}{2\alpha}})^\alpha - a \log \frac{b}{c} \\ &\leq \frac{b^{\alpha-1}}{\alpha - 1} \text{Tr} (B^{\frac{1-\alpha}{2\alpha}} A B^{\frac{1-\alpha}{2\alpha}})^\alpha, \end{aligned}$$

where $a = \text{Tr} A$, $b = \text{Tr} B$, $c = \min\{a, b\}$ and for all $\alpha \in (1, 2]$

Proof: The proof of this theorem consists of simplifying the inequality,

$$\text{Tr} A' (\log (A' + B') - \log B') \leq \frac{1}{\alpha - 1} \text{Tr} (B'^{\frac{1-\alpha}{2\alpha}} A' B'^{\frac{1-\alpha}{2\alpha}})^\alpha,$$

the details of which will not be shown here. $\square$

Challenges

So far, we have introduced a tighter bound for our original inequality. Our next objective is to relax the conditions under which the inequality holds, specifically the assumption of unital trace matrix. Although **Proposition (B)** is valid for all positive semi-definite (PSD) matrices, it includes an unwanted term, $\text{Tr} [B]^{\alpha-1}$, which we aim to eliminate.

Our desired result should be,

For positive semi-definite, self-adjoint matrices A, B

$$\text{Tr} A (\log (A + B) - \log B) \leq \frac{1}{\alpha - 1} \text{Tr} (B^{\frac{1-\alpha}{2\alpha}} A B^{\frac{1-\alpha}{2\alpha}})^\alpha$$

for all $\alpha \in (1, 2]$

However, we have not yet resolved this problem. Since the original method does not appear to be applicable in this case, we are exploring alternative approaches. The critical step that we have yet to overcome is given by

$$\text{Tr} A \frac{1}{B^s + t} D \frac{1}{B^s + t} \leq \text{Tr} A \frac{1}{B^s + t} \frac{A}{A + 1} \frac{1}{B^s + t},$$

where $s = \alpha - 1$ and $D = \frac{1}{B+1} - \frac{1}{A+B+1}$ and for all $t > 0$

Some Observations about the Inequality

To address the inequality above, we have discovered a key lemma that we believe offers a crucial step to solve the problem.

Lemma (C)

$$\lambda_i(D) \leq \lambda_i(\frac{A}{A+1}),$$

or equivalently,

$$D \leq U \frac{A}{A+1} U^*$$

where U is a unitary and $\leq$ is Loewner order.

Proof: To show this, first let $Z = 1 - \frac{1}{1+(B+1)^{-1/2}A(B+1)^{-1/2}} = \frac{(B+1)^{-1/2}A(B+1)^{-1/2}}{1+(B+1)^{-1/2}A(B+1)^{-1/2}}$. Then, $D = (B+1)^{-1/2}Z(B+1)^{-1/2}$. Also, since $\lambda_i(D) = \lambda_i(Z^{1/2}(B+1)^{-1}Z^{1/2}) \leq \lambda_i(Z)$. It suffices to show

$$\lambda_i(Z) \leq \lambda_i(\frac{A}{A+1}) = \lambda_i(1 - \frac{1}{A+1}),$$

or equivalently,

$$\lambda_i((B+1)^{-1/2}A(B+1)^{-1/2}) \leq \lambda_i(A)$$

Again, this is valid since $\lambda_i((B+1)^{-1/2}A(B+1)^{-1/2}) = \lambda_i(A^{1/2}(B+1)^{-1}A^{1/2}) \leq \lambda_i(A)$ and the desired result follows. $\square$

This type of unitary orbit relationship can also be characterized via contraction maps; see [US13]. It is clear that such a relationship alone is insufficient to establish the desired inequality. Thus, we believe that the structure of the positive semi-definite matrix

$$\frac{1}{B^s + t} A \frac{1}{B^s + t}$$

plays a crucial role in our analysis. However, there is not much reference related to this kind of inequality. Thus, it makes it a challenge to solve the problem

Conclusions

In this work, we continued our investigation into a special type of trace inequality with potential applications in quantum information theory. While certain partial results and observations have been made, the general form of the conjectured inequality remains unresolved.

Future work will focus on analyzing this inequality more deeply. We hope that understanding this structure will ultimately lead to a complete proof of the conjectured trace inequality.

References

- [MLDS⁺13] M. Müller-Lennert, F. Dupuis, Oleg Szehr, S. Fehr, and M. Tomamichel. On quantum rényi entropies: A new generalization and some properties. *Journal of Mathematical Physics*, 2013.
- [US13] Mitsuru Uchiyama and Michio Seto. Inequality between unitary orbits. *Comptes Rendus Mathématique*, 351(7):285–288, 2013.